

Simulating breeding programs: Cattle

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SCIENCE AND
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Overview

This lecture will be in 5 parts:

1. *Uses of simulating breeding programs in dairy cattle*
2. *Breeding simulation structure*
3. Selection traits and covariances
4. Systematic effects
5. Response to selection

1. Uses of simulating breeding programs in dairy cattle

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- Why we use breeding simulations in Dairy cattle?

Anticipate long-term outcomes

Predict genetic gain, inbreeding, and population trends across generations.

Compare alternative breeding strategies

Test different selection schemes, mating designs, and population structures.

Test new traits before implementation

Evaluate the potential impact of introducing traits such as health, efficiency, or resilience.

Evaluate trade-offs

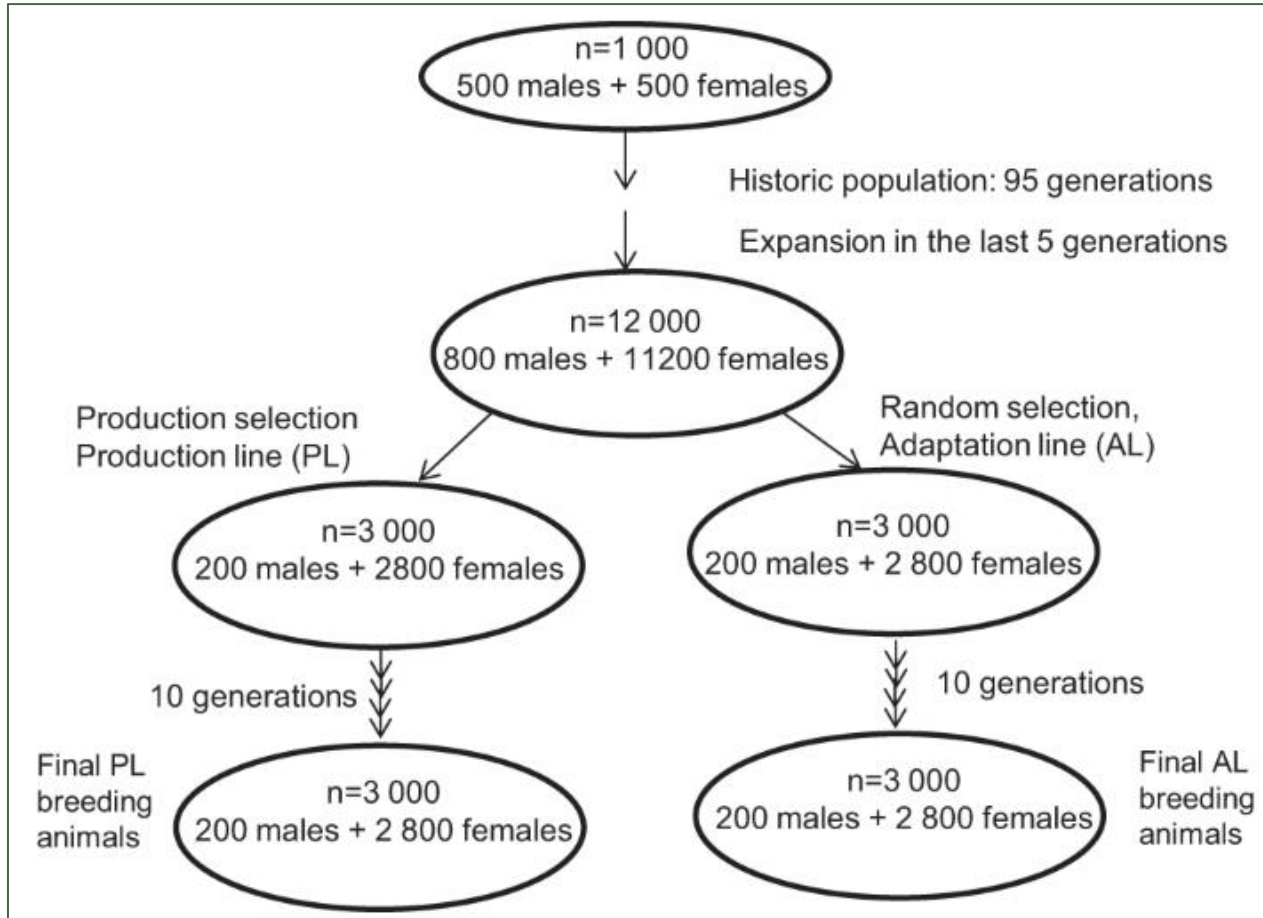
Balance production, health, disease resistance, and genetic diversity.
Identify correlated responses and/or unintended consequences of selection.

Research

Power analysis

1. Uses of simulating breeding programs in dairy cattle

- Climate adaptation in dairy cattle breeding simulations



Two cattle populations were compared:

- **Production line (PL):** high milk production, low adaptation.
- **Adaptation line (AL):** lower milk production, high adaptation.

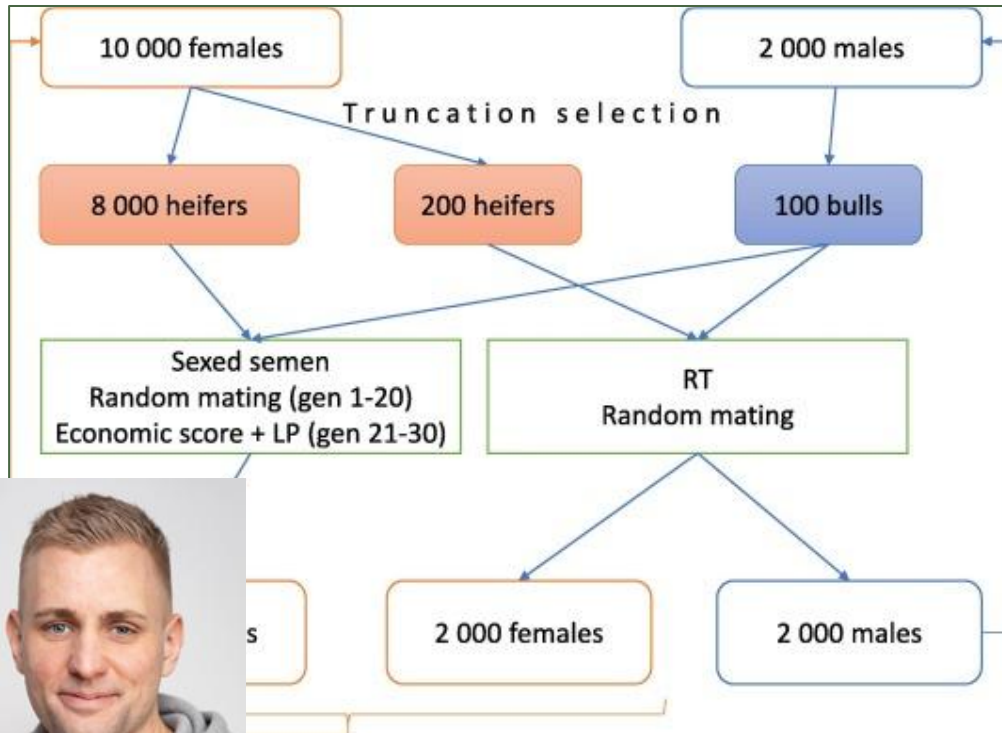
- Genomic introgression of adapted genetics into high-producing cattle helped **increase adaptation while maintaining a high production level.**
- Cross-population strategies also tended to **reduce inbreeding.**

$$h_p^2 = 0.3; h_a^2 = 0.1; r_g = -0.3$$

(Lillehammer et al., 2010)

1. Uses of simulating breeding programs in dairy cattle

- Consequences of alternative mating strategies using genomic information in dairy cattle.



- Economic mating scores** combined:
 - Total Merit Index (TMI)
 - pedigree or genomic relationships
 - penalties for recessive genetic defects
 - economic value of polledness
- Two **bull-use limits** were compared: maximum 5% or 10% of females per bull and herd.
- Including pedigree or genomic relationships **reduced inbreeding with almost no loss in genetic gain.**
- Increased the frequency of polled offspring, without substantially reducing genetic gain.
- Restricting bull use to **5%** reduced inbreeding and variation in defect-carrier frequencies.



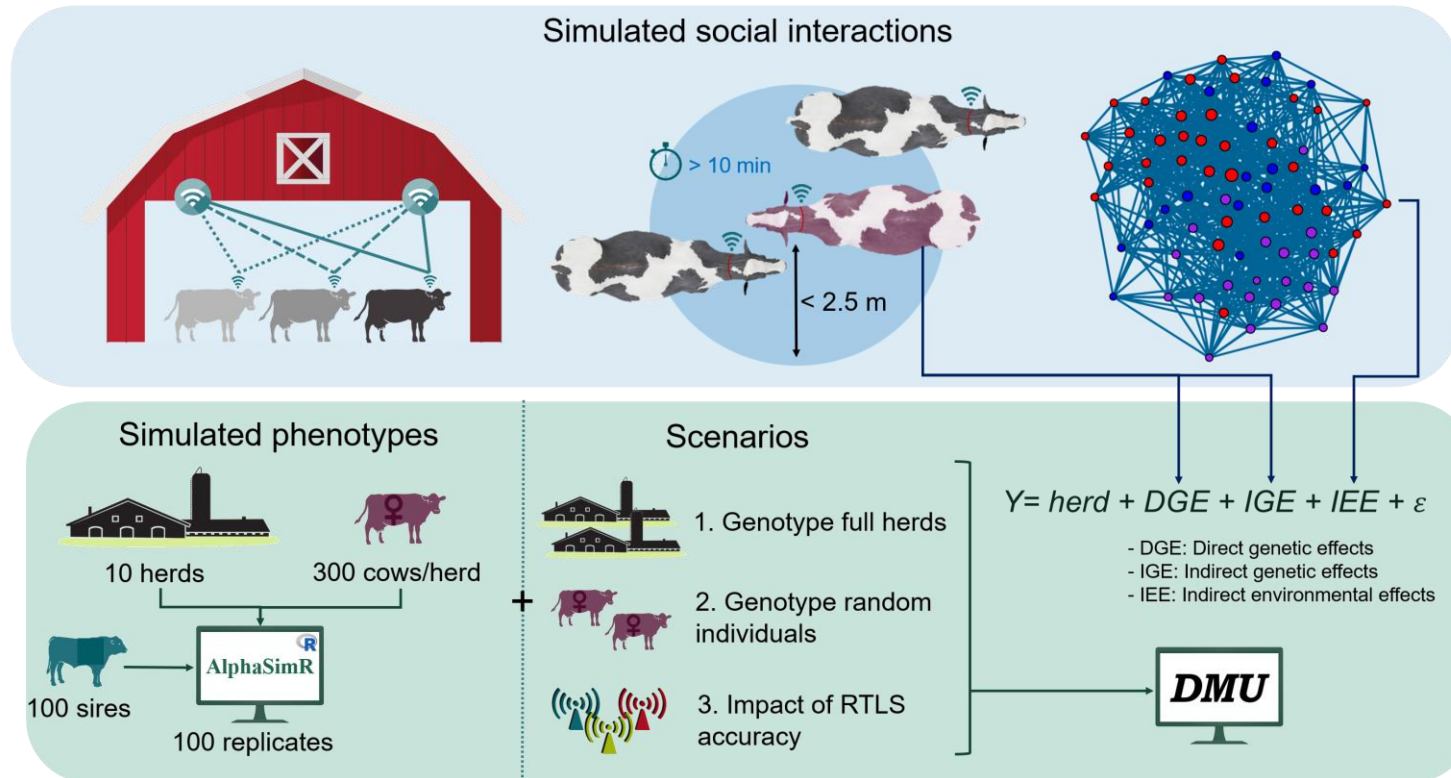
Christian Bengtsson

(Bengtsson et al., 2024)

1. Uses of simulating breeding programs in dairy cattle

- *Research:* Investigate Indirect Genetic effects

Figure 2. Social networks, phenotypes and genotypes simulation consider the direct and indirect genetic effects and indirect environmental effects from the contacted individuals.



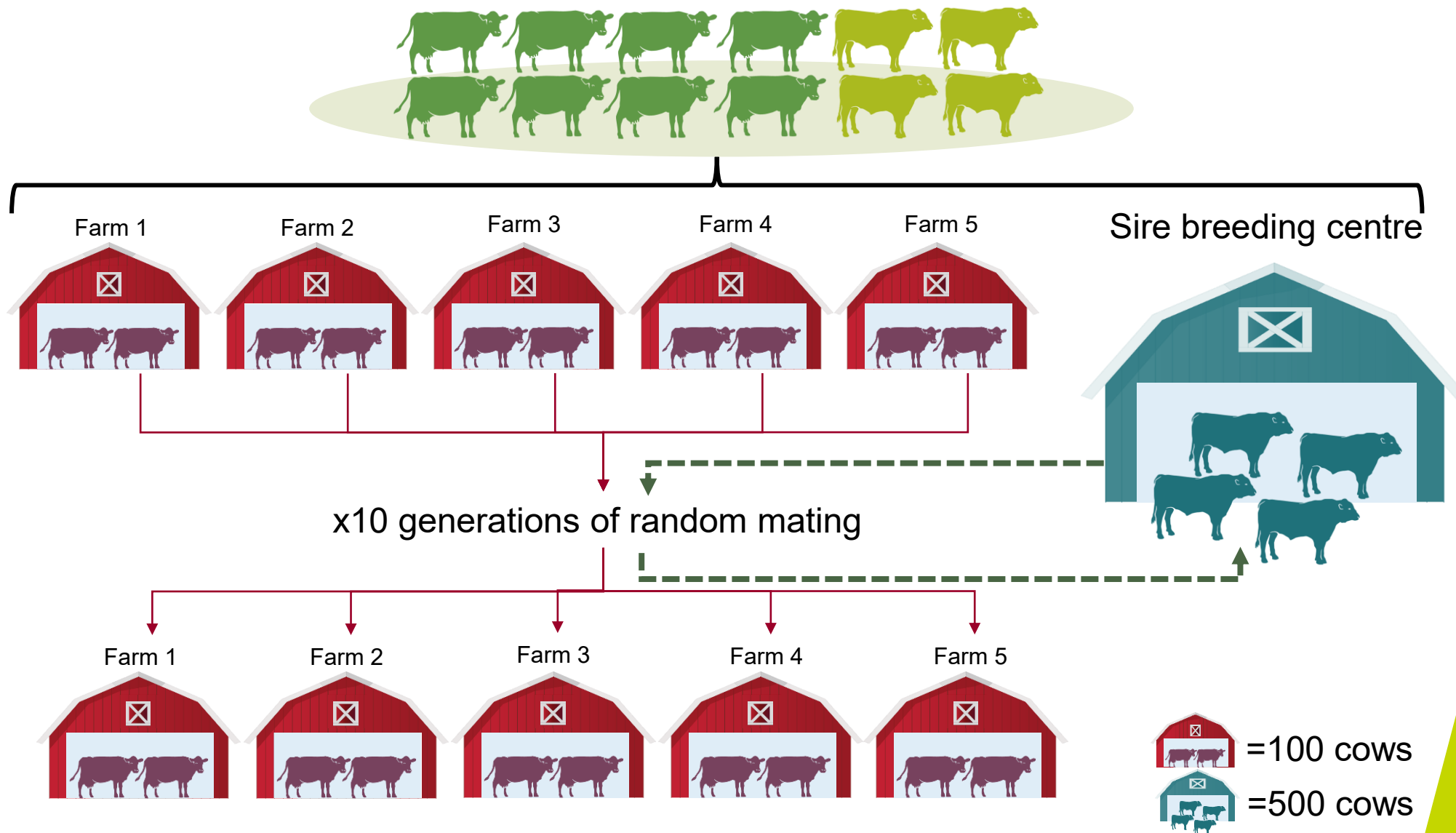
Ida Hansson

(Hansson et al., 2025)

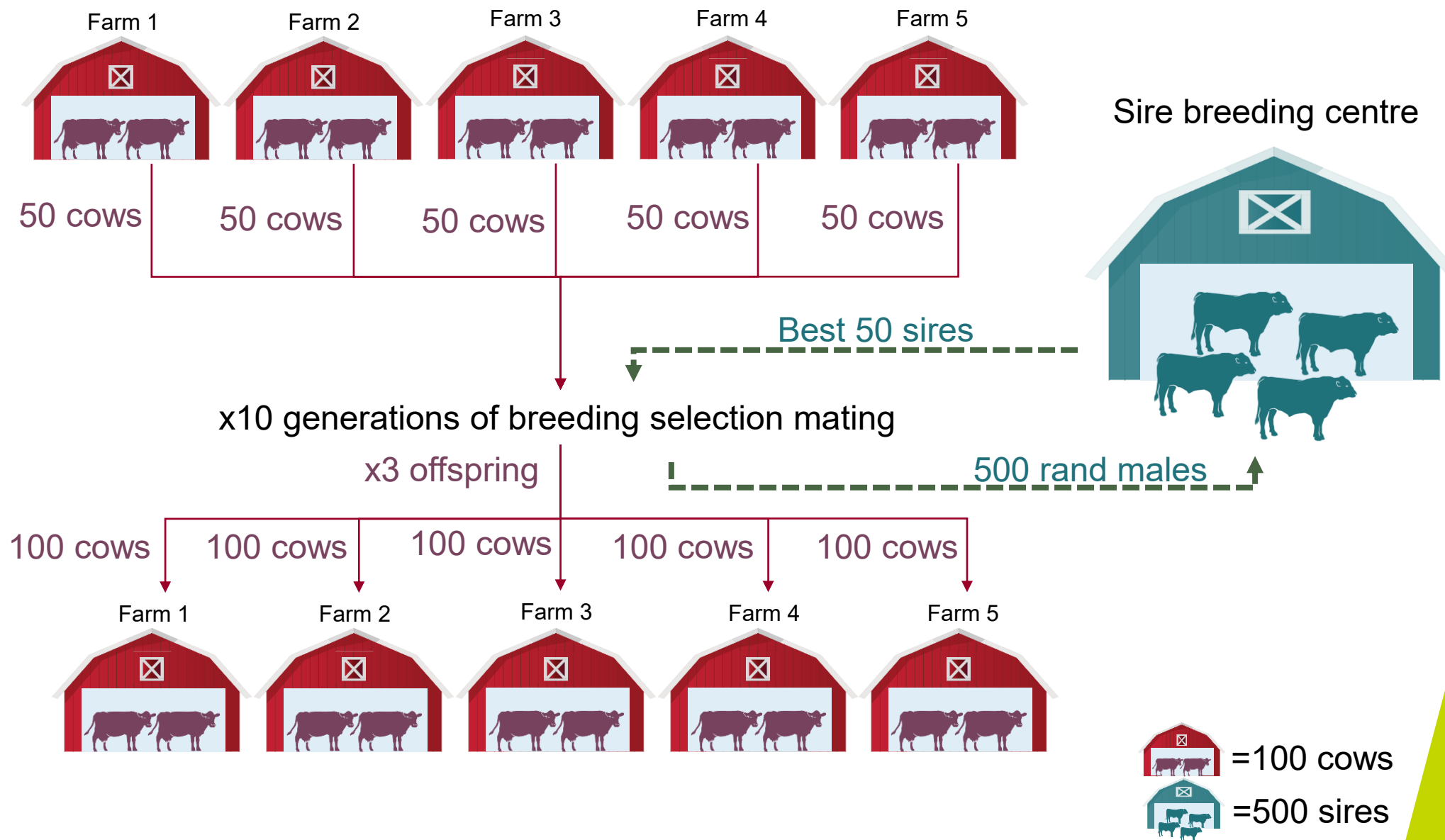
2. Breeding simulation structure

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Founder population



2. Breeding simulation structure



3. Selection traits and covariances

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Aim: estimate genetic parameters for **305-day milk production**, **lactation somatic cell score (LSCS)**, and **clinical mastitis (MAST)** during the first three lactations.

Trait	n	Phenotypic Mean	Phenotypic SD	Additive genetic SD
<i>MY305 (kg)</i>	192,652	7,575	1,552	712.2
<i>LSCS</i>	206,033	0.787	0.426	0.1524
<i>MAST</i>	221,104	0.104	0.305	0.0511

h^2/r_g	MY305 (kg)	LSCS	Clinical mastitis
MY305 (kg)	0.34	0.22	0.32
LSCS	-	0.14	0.68
Clinical mastitis	-	-	0.03



4. Systematic effects

3. Systematic effects

• Parity effect

Parity	MY305 (kg)	Effect vs. P1	LSCS	Effect vs. P1	Mastitis incidence	Effect vs. P1
1	7,575	0	0.787	0	10.4%	0
2	8,832	+1,257	0.891	+0.104	12.1%	+1.7 pp
3	9,308	+1,733	0.995	+0.208	14.9%	+4.5 pp
4*	9,308	+1,733	0.995	+0.208	14.9%	+4.5 pp

*Parity 4 was assumed equal to parity 3 for simplicity.

• Farm effect

Five farms were assigned different environmental/management.

Effects were centered approximately around zero.

Example for MY305: **-300, +200, 0, +350, and -250 kg.**

$$L = 4.5$$

(Stachowicz et al., 2011)

(Carlén et al., 2004)

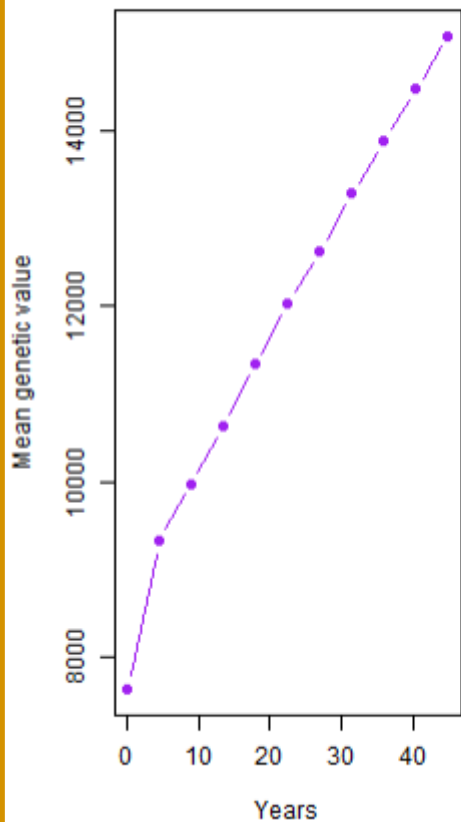
$$Y_{ijk} = \mu + GV_i + Parity_j + Farm_j + e_i$$



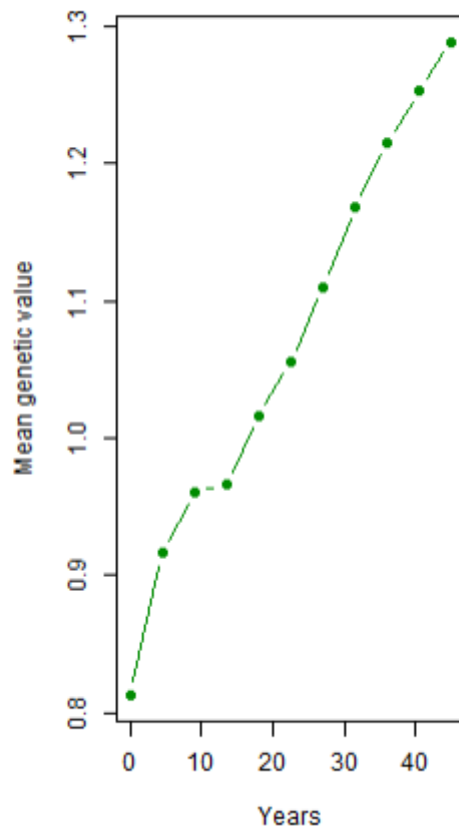
5. Response to selection

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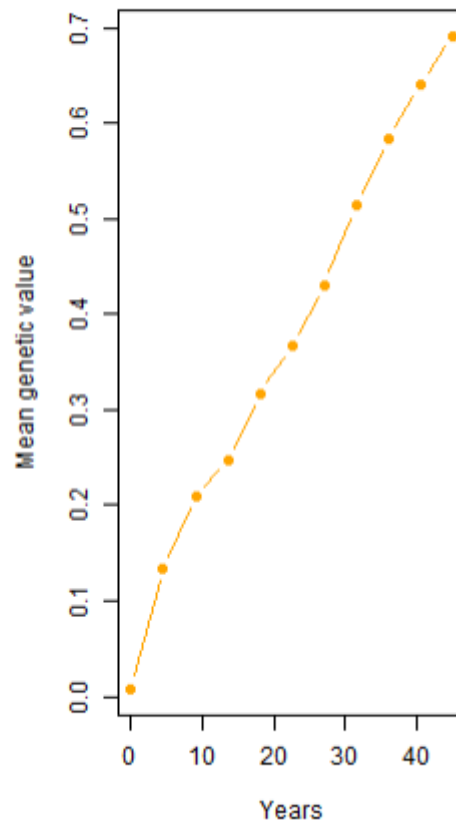
MY305



LSCS



Mastitis liability



Genetic change after 10 gen.:

MY305: + 7427.3 kg

LSCS: + 0.476

Mastitis liability: + 0.684

3. Selection traits and covariances

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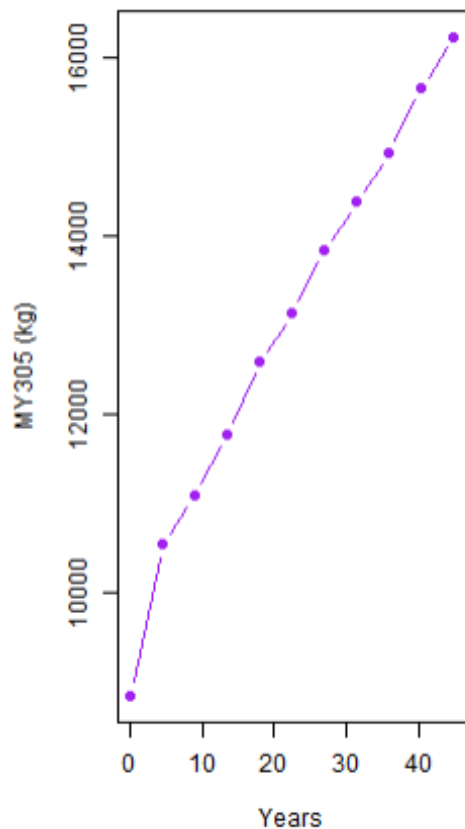
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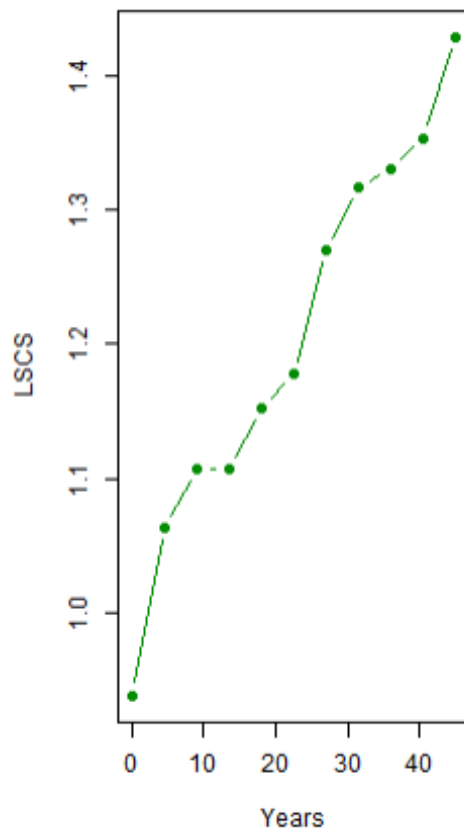


5. Response to selection

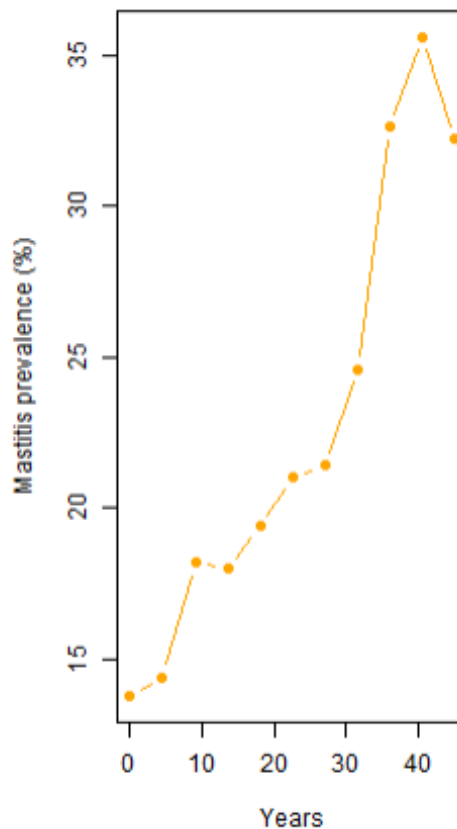
Milk production



Somatic cell score



Clinical mastitis



Genetic change after 10 gen.:

MY305: + 7427.3 kg

LSCS: + 0.476

Mastitis liability: + 0.684

Phenotypic change after 10 gen.:

MY305: + 7391.9 kg

LSCS: + 0.491

Mastitis prevalence: + 18.4 pp

- *Essential*

- Gaynor, R.C., Gorjanc, G., & Hickey, J.M. 2021. AlphaSimR: an R package for breeding program simulations. *G3*. <https://doi.org/10.1093/g3journal/jkaa017>
- Lillehammer, M., Meuwissen, T.H.E., & Sonesson, A.K. 2011. A comparison of dairy cattle breeding designs that use genomic selection. *Journal of Dairy Science*. <https://doi.org/10.3168/jds.2010-3518>
- Bengtsson, C., Stålhammar, H., Thomasen, J.R., Fikse, W.F., Strandberg, E., Eriksson, S., & Johnsson, M. 2025. Simulation of long-term impact of dairy cattle mating programmes using genomic information at the herd level. *Animal*. <https://doi.org/10.1016/j.animal.2025.101498>
- Hansson, I., Bijma, P., Fikse, F., & Rönnegård, L. 2025. Towards assessing indirect genetic effects in dairy cattle. *Genetics Selection Evolution*. <https://doi.org/10.1186/s12711-025-00988-w>

- *Further*

- Stachowicz, K., Sargolzaei, M., Miglior, F., Schenkel, F.S. 2011. Rates of inbreeding and genetic diversity in Canadian Holstein and Jersey cattle. *Journal of Dairy Science*. <https://doi.org/10.3168/jds.2010-3308>
- Carlén, E., Strandberg, E. & Roth, A. (2004). Genetic parameters for clinical mastitis, somatic cell score, and production in the first three lactations of Swedish Holstein cows. *Journal of Dairy Science*. [https://doi.org/10.3168/jds.S0022-0302\(04\)73439-6](https://doi.org/10.3168/jds.S0022-0302(04)73439-6)
- Medrano, J.F., Ahmadi, A., & Casellas, J. 2010. Dairy Cattle Breeding Simulation Program: A simulation program to teach animal breeding principles and practices. *Journal of Dairy Science*. <https://doi.org/10.3168/jds.2009-2640>
- Strandén, I., Kantanen, J., Russo, I.-R.M., Orozco-terWengel, P., Bruford, M.W., & Climgen Consortium. 2019. Genomic selection strategies for breeding adaptation and production in dairy cattle under climate change. *Heredity*. <https://doi.org/10.1038/s41437-019-0207-1>

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